

ELECTRICITY AND WATER DESALINATION: SEPARATE SITES OFFER VALUE

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Abstract. For the fuel type and water situation in the Middle East, the case is strong for the use of combined cycle technology for power generation and reverse osmosis for potable water production, where each are sited for their maximum economic benefit and interconnected by electric power transmission. Because of the fuel efficiency of Combined Cycle generation technology, its use of liquid/gas fuels and its low need for cooling water, it can be optimized for cost away from cities. Conversely, water desalination by reverse osmosis can be sited in optimal locations to take advantage of its modularity and to minimize water pipeline needs. Electric power transmission provides an inexpensive and flexible means to connect these two technologies. Together these technologies may offer an overall minimum cost approach, better than the combining of electric power and water desalination at one location, where power to water ratios must be fixed, independent of need, for optimum efficiency. The use of reverse osmosis with power generation has other, important ancillary benefits over using distillation and power combinations. These advantages include abatement of environmental pollution, delivery of potable water at reasonable drinking temperatures, lower total energy consumption, more efficient land use and less demanding operator skills.

Keywords: Desalination, reverse osmosis, multi-stage flash, combined cycle power plant, electric transmission, plant siting, steam power plant, multiple effect distillation, cost comparison

1. Introduction

Two basic staples of modern society are water and electricity. Upon these two commodities humankind has established an industrial society that continues to thrive, even as the number of people climbs.

But because these two essential staples are used together to feed our modern, industrial world does not mean they must be produced together. Indeed, it is the thesis of this paper that there are significant opportunities for both electricity generation and water production by desalinization in which the processes should be separated for best economy. The separation of production facilities should be tested for the use of the most economically efficient combination of both electric and water production facilities and their respective delivery systems. The economics should consider not only respective cycle efficiencies, but also relative environmental effects, land values, joint facilities use and fuel/water/power delivery requirements.

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2. Electric Power Efficiency Improvements

The continued increase in efficiency of combustion turbine-based electric power production facilities, especially in combined cycle operation, suggests that a move away from boiler-based electric power production with steam extraction for water production may be in order. This is true whether the combustion turbine waste heat is captured by a heat recovery steam generator (HRSG) for use in power generation in conjunction with reverse osmosis (RO), or in supplying a desalination facility. The move to separate power production and desalination by using combined cycle power generation and RO would also have the important benefit of freeing up the problem of siting from the double constraint of both fuel sourcing and seawater access. The low electric power consumption of reverse osmosis water desalination further suggests that the capital requirements for the combined electric and water transmission and their inherent transmission power costs could be reduced by shifting the burden of transmission to the more flexible electric power network. The locating of reverse osmosis facilities, which have very little air and water environmental impact, near seawater and population centers would provide a potential improvement in city quality of life over combination facilities while keeping highly efficient but pollutant-emitting fuel burning at more remote sites.

A subtle but valuable advantage of separating water from electricity production is the ability to size the individual production facilities to suit their specific needs, not having to compromise the technical efficiency of a single plant for both purposes on the one hand, or the right balance of water needs and electricity demands by the community on the other.

3. Technology Overview

Steam boiler-based thermal desalination technologies include multi-stage flash (MSF), multiple effect distillation (MED), with mechanical vapor compression (MVC) and thermal vapor compression (TVC). The steam was formerly derived from conventional boiler and steam turbine power generation cycles with extraction. Presently, the most efficient way to provide the steam needs for MSF or MED is through steam developed by a heat recovery steam generator at the exhaust of a combustion turbine power generation cycle. The steam from the HRSG can be used with an extraction turbine or as supply to a desalination process exclusively. MSF has traditionally been the most widely used technology, though MED is now emerging with an increased market share. All of these systems essentially work the same way, by using steam from a boiler to vaporize and then condense seawater. A typical single-purpose distillation plant might be comprised of an oil-fired steam plant or a combined cycle power generation island, and a distillation/condensation unit within the same facility. A single-purpose distillation plant directs all of the steam generated by the boiler to the distillation unit, while a dual-purpose plant diverts only part of the steam and leaves the rest for power generation.

Most of the largest MSF and MED plants are dual-use, as the design offers better efficiency than single-use plants, of course, assuming that the power generation can be effectively used.

A reverse osmosis configuration, by comparison, separates the power generation and water desalination processes. This allows the two functions to be performed in separate locations, with each function using the technology optimal for the task. Power can be generated using a modern, efficient technology such as combined cycle (CC) gas-fired plants, while the desalination is performed using RO, a clean safe process, close to a large population center.

The use of a combined cycle plant would provide economic advantages over a traditional steam boiler and turbine arrangement, both in capital construction costs and in operation and maintenance costs. Using recent estimates, a modern 250 MW gas-fired CC plant would cost approximately 40% less to operate in the Middle East than a traditional steam plant of comparable size. The total economic efficiency arises from lower construction capital costs and lower operation and maintenance costs, as well as a lower heat rate. In addition, the CC plant also enjoys the advantages of lower emissions and shorter plant installation time. [See Reference 1.]

These capital cost advantages found in a CC facility used only for power generation would also apply to the combustion turbine and HRSG portion of a facility supplying steam to a single or dual-purpose desalination plant. However, the operating efficiencies would be comparable to a conventional steam power plant. Therefore, this use would provide economics that fall between the traditional plant and the power only CC case modeled later in this paper, but not provide the advantages of optimizing separate siting and sizing of power and desalination facilities.

TABLE IV
Comparison of Desalination Plant Capital Costs
For 6 MGD (22,700 CMD) RO and MSF plants on the Arabian Gulf*

	RO		MSF		MED	
	US \$(000s)	Percent	US \$(000s)	Percent	US \$(000s)	Percent
Direct Capital Costs						
Permeators, Installed	6,000	15%				
Process Equipment, Installed	16,500	40%	33,500	57%	41,500	63%
Site Development	1,000	2%	1,000	2%	1,000	2%
Intake and Outfall Systems	7,000	17%	9,065	15%	6,593	10%
Total Direct Costs	30,500	74%	43,565	74%	49,093	74%
Indirect Capital Costs						
Interest During Construction	1,525	4%	2,178	4%	2,455	4%
Contingency	2,440	6%	3,485	6%	3,927	6%
A&E Fees, Project Mgmt.	5,185	13%	7,406	13%	8,346	13%
Working Capital	1,525	4%	2,178	4%	2,455	4%
Total Indirect Costs	10,675	26%	15,247	26%	17,183	26%
Total Capital Cost	41,175	100%	58,812	100%	66,276	100%
US \$/GPD	\$6.86		\$9.80		\$11.05	
US \$/CMD	\$1,813		\$2,590		\$2,918	

* Note - Values for MSF and MED plants are scaled to 6MGD from 25 MGD plants.

4. Capital and Operational Costs

The capital cost of a 6 million gallons/day (MGD) [22,700 cubic meters/day (CMD)] RO plant is estimated to be approximately 2/3rds the cost of an MSF or MED plant of the same size. (See Table IV above, Comparison of Desalination Plant Capital Cost.)

Non-fuel operation and maintenance expenses are also estimated to be lower for the RO plant. Conversely, the station service power consumption for an RO plant in a Middle East application would be approximately 20-25 KWh per 1000 gallons (5.3-6.7 kWh/M³), while a single-use MSF or MED plant would require approximately 55% or 32% of that, respectively. (See Table V, Comparison of Desalination Energy Requirements.)

TABLE V

**Comparison of Desalination Energy Requirements
For 6 MGD (22,700 CMD) RO and MSF plants on the Arabian Gulf**

	RO		MSF		MED	
	KWH/KGals	KWH/CM	KWH/KGals	KWH/CM	KWH/KGals	KWH/CM
Seawater Supply	1.5200	0.4016	3.2625	0.8620	1.5833	0.4183
Booster Pump	1.5200	0.4016				
Chemical Dosing Pumps	0.0700	0.0185				
High Pressure Pumps	34.0500	8.9960				
Energy Recovery	-13.3600	-3.5297				
Degasser Blower	0.0700	0.0185				
Product	0.6100	0.1612				
Recirculating Pump			10.8750	2.8732	6.5250	1.7239
Distillate Pump			0.0400	0.0106	0.0400	0.0106
Plant Services	1.0000	0.2642				
Total	25.4800	6.7318	14.1775	3.7457	8.1483	2.1528

It should be noted that most dual-use MSF plants use low-pressure exhaust steam from the electrical generator and hence have a lower energy requirement. This is particularly true of MED technology, which utilizes lower temperature steam than that required by MSF. The use of waste heat, however, takes away some power that can be generated from the residual steam. This electric power loss must be included in any process energy economic comparisons, including the capital cost to achieve the same electrical output while supplying adequate steam for the desalination process.

Operational costs of the various desalination technology plants are comprised of labor, maintenance and parts, chemicals and consumables, and in the case of RO, membrane replacement. General maintenance and parts cost is higher for RO than MSF/MED, but labor cost is lower, as constant monitoring of the equipment is not required. However, RO technology requires periodic replacement of the membrane. The membranes are guaranteed for 5 years, but experience has shown they may last 10 years or more. Under a reasonable 8-year life estimate, membrane replacement makes up approximately 21% of total O&M expenses, excluding capital carrying costs, and 10% of total costs,

including capital. (See Table VI, Comparison of Total Water Costs, which is developed from information in Tables 1, 2, and 3, which are located at the end of the article.)

TABLE VI
Comparison of Total Water Costs
For 6 MGD (22,700 CMD) RO and MSF plants on the Arabian Gulf

Assumptions

Financial Assumptions:

Cost of Capital	8.0%
Plant Amortization Period	30
Fixed Charge Rate	8.8627%

Conversion Factors:

Gallons → Cubic Meters	0.00379
Cubic Meters → Gallons	264.2

Plant Assumptions:	Units	RO	MSF	MED
Plant Capacity	MGals/Day	6	6	6
Desalination Capacity Factor		90%	90%	90%
Annual Production	MGals	1,971	1,971	1,971
Process Power Used	KWH/KGals	25.48	14.18	8.15
Cost of Electric Power	\$/MWH	\$19.87	\$23.37	\$23.37
Cost of Power after 2.5% Transm. Losses	\$/MWH	\$20.37	\$23.37	\$23.37
Average Membrane Life	Years	8.0		
Steam Equiv. Power Consumption	KWH/KGals		0.00	0.00
Total Equiv. Power Consumption	KWH/KGals	25.48	14.18	8.15
Plant Capital Cost	\$000	\$41,175	\$58,812	66,276

Annual Operating Costs

	RO		MSF		MED	
	US \$'000s	Percent	US \$'000s	Percent	US \$'000s	Percent
Process Electric Power	\$998	14%	\$0	0%	\$0	0%
Steam	—		\$0	0%	\$0	0%
Consumables, Chemicals	341	5%	410	6%	200	3%
Maintenance & Parts	915	13%	500	7%	500	7%
Supervision & Labor	283	4%	800	12%	800	11%
Membrane Replacement	700	10%	—		—	
Subtotal	3,237	47%	1,710	25%	1,500	20%
Fixed Charges	3,657	53%	5,224	75%	5,887	80%
Total Annual Costs	<u>\$6,894</u>	<u>100%</u>	<u>6,934</u>	<u>100%</u>	<u>7,387</u>	<u>100%</u>

Net Water Cost:

US \$/1000 Gals	\$3.50	\$3.52	\$3.75
US \$/Cubic Meter	\$0.92	\$0.93	\$0.99

In total, the production cost of comparable RO, MSF, and MED facilities would typically be about \$3.50-\$4.00 per 1000 gallons of distilled product, respectively (\$0.90 to \$1.00 per M³).

4.1. Transmission Costs

A hypothetical RO/CC configuration would use a combined cycle plant located away from populated areas, for instance in the desert, transmitting the electric power through the electric transmission network to a RO plant located near a major city. The incremental transmission to deliver the incremental power for in-city RO operation is not likely to be more than that required to interconnect reliably the electric power plant itself. However, this type of use will cause incremental power transmission losses, which normally average about 2 - 3% of increased power delivery on the transmission network (the Comparison of Total Water Costs , Table VI, uses a loss factor of 2.5%). On the other hand, locating a MSF plant away from populated areas will also incur capital and operating costs for transmission of desalinated water.

4.2. Fuel & Water Supply Issues

One concern with MSF technology for the Middle East is that it requires large quantities of water for cooling as well as the desalination process. As such, the power production facility would need to be located near the coast, and either shipping or pipeline facilities would be required to fuel the plant. A 250 MW CC plant with dry cooling for its steam cycle, by comparison, requires much less water, on the order of 5,000 to 8,000 gallons per day. Hence, by trucking or piping in the water needed, the plant could be located in a desert at the most useful fuel supply and electric transmission location, perhaps close to a source of natural gas, for example.

4.3. Pollution Issues

A major argument in favor of separating the power generation and desalination functions is the concern over pollution from electric plant emissions. Many Middle Eastern countries have populations that are concentrated in small areas, and the effects of pollution from additional nearby power plants are certain to be detrimental. Typically for gas or oil fired electric power plants, the pollution most encountered is ground level ozone that results from the chemical interaction of nitrogen oxides (power plant emissions) and volatile organic compounds (automobile exhausts) in the presence of sunlight. This is the classic "smog," which is the mark of many modern cities. Although it is difficult to place an objective estimate on the value of placing a power plant away from populated areas in the Middle East, it is certainly a factor that merits consideration.

4.4. Combined Plant Versus Separate Plant Size Efficiency

Another concern with the distillation technologies is the inefficiency of mismatched scale economies. Steam plants tend to be large, with 500-600 MW being a typical figure for a scale-efficient plant. At a typical 8:1 or 10:1 power-to-water ratio for

distillation technology, this results in a 50-75 MGD (190,000 - 280,000 CMD) desalination facility. A 250-300 MW combined cycle plant with MSF or MED desalination would have a water supply range of 20 to 30 MGD (76,000-114,000 CMD). This is in comparison to a typical desalination requirement of 15-50 MGD (57,000 - 190,000 CMD) for most population centers under consideration. As such, a steam boiler-based MSF plant located near an urban area might have considerable excess desalination capacity. Of course, water could be transported to other markets, but this would incur additional transmission costs. A CC plant with MSF or MED would be a much better fit to typical needs.

In comparison, the use of a local RO plant would best match production to local consumption requirements. A scale-efficient RO plant can be as small as 6 MGD (22,700 CMD), which would better serve the needs of many small to medium sized cities. At the same time, the generation facilities could also be sized to achieve scale economies, whatever the technology involved for power production.

There will be some cases in which other considerations may favor the use of distillation technology over RO. For instance, an area might already have surplus steam available from an existing power plant. In such cases, it would be difficult to justify installation of an RO facility when a distillation unit could be retrofitted to the existing power installation. Conversely, the more likely situation will be that in which sufficient electric power generation is in place, but there is inadequate water supply. Addition of an RO facility will be very cost effective in this situation.

5. Cost Evaluation

Ultimately, the cost and power issues need to be examined together within the framework of a fully integrated analysis. In its past experience in the Middle East, Stone & Webster has utilized tools such as EPRI's EGEAS(tm) electric power planning software (for which Stone & Webster is the developer and commercializer for EPRI) to model and optimize the economies of various technologies. EGEAS is a state-of-the-art modular production costing, generation expansion software package developed under EPRI sponsorship for use by utility planners to evaluate integrated resource plans, independent power producers, avoided costs and plant life management programs. Only through such a formalized evaluation process can the merits of various scenarios be accurately determined. DuPont's expertise as the world's foremost and largest membrane supplier offers significant assistance in design, operations and maintenance of RO plants.

In order to illustrate some of the issues just presented, we have performed a basic spreadsheet cost calculation comparing a hypothetical RO plant versus comparable MSF and MED plants, all sited in the Middle East. The RO configuration uses the DuPont Permasep(tm) 6880T membrane paired with a 250 MW CC power plant. The MSF and MED packages are assumed to be powered by dual-purpose 300 MW steam plants.

Some assumptions had to be made in order to streamline the calculations for the purposes of this paper. For comparability, the capital and operating cost figures for all

three technologies were based on a capacity of 6 MGD (22,700 CMD). In reality, the MSF and MED plants would have larger capacity matched to the accompanying power generation facilities. Under the assumptions made in this analysis, such capacity would be around 25-35 MGD (94,600 -132,500 CMD).

Total Water Cost	RO	MSF	MED
Cost, US \$/1000 Gals Produced	\$3.50	\$3.52	\$3.75
Cost, US \$/M ³ Produced	\$0.92	\$0.93	\$0.99

The analysis indicates that under the hypothetical scenario, RO technology enjoys a small cost advantage over MSF, and a slightly larger advantage over MED. (See Table VI, Comparison of Total Water Costs.) Closer inspection of the calculations indicates that the water cost for both MSF and MED are burdened by the higher capital cost of those technologies. In addition, MSF and MED also incur substantial costs for the steam that is bled from the generator. This shows up in the analysis in the form of higher capital costs for the power plant and a penalty on heat rate, but no other steam costs. Charges were taken for additional auxiliary power needs for the desalination plant that were not included in the CC plant net plant calculation.

In addition to the cost calculations, there are also other assumptions that bear on the analysis results. They include the plant's load factor, RO membrane replacement rate, and the cost of power, all of which are subject to change based on local conditions. Still, the example illustrates the kind of analysis that can be conducted to evaluate the suitability of a particular technology to local needs.

6. Conclusion

In comparing the advantages of RO desalination with combined cycle electric production versus the same power technology with MSF or MED desalination, we have examined several factors both tangible and intangible, which need to be taken into account:

- The efficiency of more modern generation technologies such as combined cycle plants regardless of how they are coordinated with desalination.
- The choice of single-use vs. dual-use thermal distillation facilities. Dual-use facilities can lower the power cost of the distillation process markedly over single-use plants.
- The capital and operational costs of both thermal distillation and reverse osmosis technologies.
- Transmission costs and losses associated with transmitting both power and water.
- Fuel and water supply issues related to siting power plants.
- The societal cost of emissions, and the desirability of locating a generation plant away from populated areas.

- Efficiencies involved in sizing power plant and desalination capacity according to local requirements and scale efficiencies.

Only when all of these factors are evaluated simultaneously can a technology be selected as the best choice for any particular application. Therefore, we believe reverse osmosis in conjunction with efficient remote power generation offers enough benefits for the Middle East region that it merits further site-specific investigation.

TABLE I
BASIS OF DESIGN
LOCATION ARABIAN GULF

Plant Capacity - 6 MGD, 22,700 CMD	Conversion - 35%
DuPont Permeator Model - B-10 TWIN™, 6880T	Single Pass RO System.
Product Quality - Less than 500 mg/l delivered	Term - 5 years
Open Sea Feedwater Intake	RO Feed Pressure - 1200 psig maximum
Operating Hours per Year - 7884	Efficiency of Motors - 92%
Annual Capacity - 1971M Gallons, 7460 KM ³	Efficiency of High Pressure Pump - 80%
Number of Trains - 4	Efficiency of Energy Recovery Device - 82%
Number of Blocks per Train - 2	Energy Cost - US \$0.05 per kWh
Capacity per Train - 1500 KGD, 5675 CMD	Product Pressure - 10 psig
Feed Water Composition -Standard seawater at 45,000 mg/l.	RO Feed pH - 8.3
Temperature Range - 17°-35°C.	Piping:
	Seawater - SMO 254 or equivalent
	Product - 316L or plastic

TABLE II
Seawater RO Total Water Costs
For A 6 MGD (22,700 CMD) Plant on the Arabian Gulf

Annual Operating Costs		
	<u>US \$(000s)</u>	<u>Percent</u>
Electric Power at \$0.019875/KWH	998	14%
Consumables, Chemicals	341	5%
Maintenance and Parts ¹	915	13%
Supervision and Labor	283	4%
Membrane Replacement ²	700	10%
Total Direct Costs	<u>3,237</u>	<u>47%</u>
US \$/1000 Gallons	\$1.64	
US \$/CM	\$0.43	
Fixed Charge (Amortization) at 8%, 30 Years	<u>3,657</u>	<u>53%</u>
Total Annual Cost	<u><u>6,894</u></u>	<u><u>100%</u></u>
US \$/1000 Gallons	\$3.50	
US \$/CM	\$0.92	

Notes:

1. 3% of Direct Capital Costs
2. Assumed 12% per year replacement rate

TABLE III
Comparison of Generation Technologies
For Combined Cycle Power Plants with Extraction for MSF/MED or Standalone

Assumptions			
Technology Assumptions			
CC, 250 MW Net Output			
		w/ Extraction	w/o Extraction
Capital Installation Cost	\$/KW	800	663
Fixed O&M	\$/KW-Year	13.09	13.09
Variable O&M	\$/MWH	0.71	0.71
Economic Assumptions			
Cost of Fuel - \$/MM Btu			\$1.00
Cost of Capital			8.0%

Results at 75% Capacity Factor		
CC, 250 MW Net Output		
	w/ Extraction	w/o Extraction
Levelized Generation Cost - \$/MWH	\$23.372	\$19.875

Plant Type: **COMBINED CYCLE w/Extraction for MSF or MED**

ISO Capacity (MW)	253	Book Life	30
Derated Capacity (MW)	197	Capital Recovery Factor	0.08883
Capital Cost (\$/KW)	\$800.12	Annual Fixed Cost	\$14,016
Variable Cost (\$/MWH)	\$0.71		
Fuel Cost (\$/MMBTU)	\$1.00	Average Heat Rate (BTU/KWH)	9,852
Fixed O&M (\$/KW-Yr)	\$13.09	Fuel Cost (\$/MWH)	\$9.85

Heat Rate Blocks	100%	75%	50%	25%	10%
(At 105° Ambient Temp)	9,602	10,110	11,457	14,988	15,889

Capacity Factor	Annual Fixed O&M (\$'000'S)	Annual Energy (MWH)	Annual Variable Cost (\$'000'S)	Annual Fuel Cost (\$'000'S)	Annual Total Cost (\$'000s)	Levelized Power Cost \$/MWh
70%	2,581	1,209,296	859	11,914	29,371	24.287
75%	2,581	1,295,674	920	12,765	30,283	23.372
80%	2,581	1,382,052	981	13,616	31,195	22.572
85%	2,581	1,468,431	1,043	14,467	32,108	21.865

Plant Type: **COMBINED CYCLE 250MW**

ISO Capacity (MW)	253	Book Life	30
Derated Capacity (MW)	197	Capital Recovery Factor	0.08883
Capital Cost (\$/KW)	\$662.96	Annual Fixed Cost	\$11,614
Variable Cost (\$/MWH)	\$0.71		
Fuel Cost (\$/MMBTU)	\$1.00	Average Heat Rate (BTU/KWH)	8,209
Fixed O&M (\$/KW-Yr)	\$13.09	Fuel Cost (\$/MWH)	\$8.21

Heat Rate Blocks	100%	75%	50%	25%	10%
(At 105° Ambient Temp)	7,959	8,380	9,496	12,423	13,170

Capacity Factor	Annual Fixed O&M (\$'000'S)	Annual Energy (MWH)	Annual Variable Cost (\$'000'S)	Annual Fuel Cost (\$'000'S)	Annual Total Cost (\$'000s)	Levelized Power Cost \$/MWh
70%	2,581	1,209,296	859	9,927	24,981	20.657
75%	2,581	1,295,674	920	10,636	25,751	19.875
80%	2,581	1,382,052	981	11,345	26,522	19.190
85%	2,581	1,468,431	1,043	12,054	27,292	18.586

References

Stone & Webster Engineering Corporation: 1993, *Development of Engineering, Cost, and Performance Data for Generation Supply Options for New England*, 541 pages.

Glossary

Combined Cycle (CC) - A generation technology based on combustion turbines that uses heat recovery steam generation to recover waste heat from the exhaust and uses that waste heat to make steam to drive a steam turbine generator. Cycle efficiencies above 50% and relatively low capital costs characterize this technology.

EGEAS - An integrated resource planning (IRP) model developed and maintained for EPRI and its members by Stone & Webster. The model is a world-wide leading innovation for performing production cost and capital cost optimization analysis for electric power supply. As a major part of the EPRI IRP Workstation, EGEAS integrates the cost analysis for supply-side (generator) resources with demand-side management processes and works with financial planning and risk assessment tools to support power supply business decisions.

Multi-Effect Distillation (MED) - A thermal desalination process in which evaporation takes place as a thin film of feed water passes over a heat transfer surface (outside of horizontal tubes - HTE or inside of vertical tubes - VTE). Vapor formed in the last effect is recompressed thermally (TVC) or mechanically (MVC).

Multi-stage Flash Distillation (MSF) - Also a thermal desalination process in which a stream of brine flows through the bottom of up to 25 chambers (stages). The pressure in each chamber is maintained at a lower level than the saturation vapor pressure of the water and a proportion of it “flashes” into steam and is then condensed.

Reverse Osmosis (RO) - A pressure greater than the osmotic pressure of the feed water is applied to one side of a semi-permeable membrane, producing a flow of desalinated water through the membrane.